

Circuit Analysis (Lab # 2)

Experiment 2: Measurements of Waveforms and Fundamental Quantities, Basic resistive circuits.

Name: Muhammad Ahsan Ali

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Grade and Signature:

CLO3	Make measurements in their own constructed electric circuits.			
Psychomotor/Affective	Level1 (1)	Level2 (2-3)	Level3 (4-5)	Level4 (6-7)
Report Marks (3)			Total marks (10)	

Objectives: This experiment will provide exposure to measurements of waveforms and Fundamental Quantities, Basic resistive circuits. A primary purpose of this lab course is for you to master the use of electronic test equipment and measurement of fundamental quantities.

Equipment required:

Power Supply, Breadboard, Oscilloscope, Signal Generator Connecting wires, Resistances, Multimeter. **Procedure**

Experiment # 1

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'A' shown below in Figure 1.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 100 Hz and peak-to-peak voltage of 5V.
4. Set up one probe across input signal, and another across R1 and R2 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

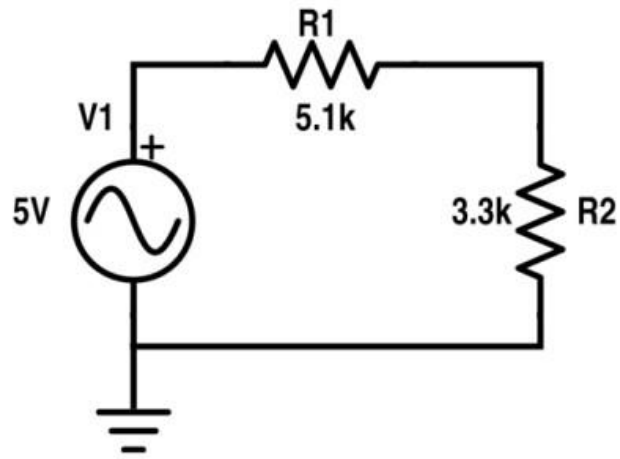


Figure 1 Circuit 'A'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	5	1.07	3.04
R2	5	0.69	1.96

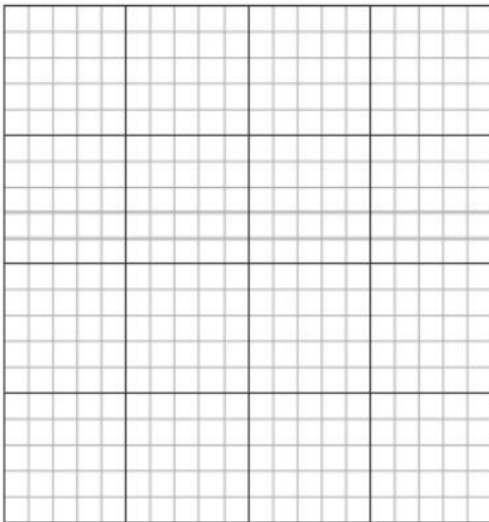


Figure 2 Input Voltage Signal

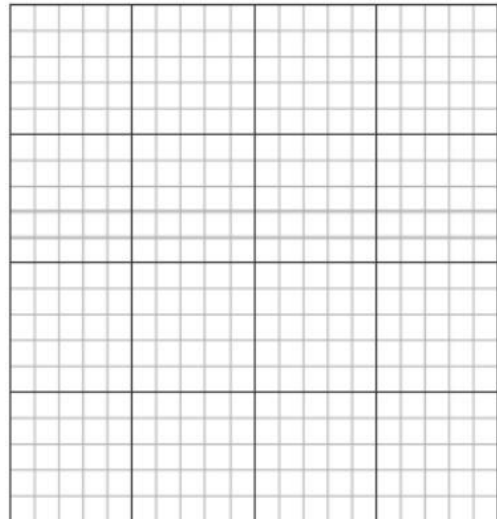


Figure 3 Voltage Signal across R1

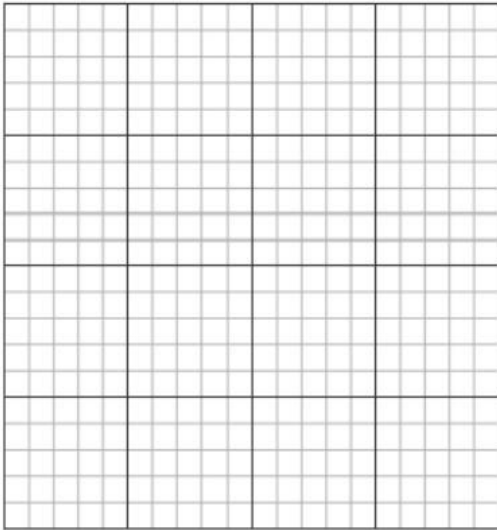


Figure 5 Voltage Signal across R2

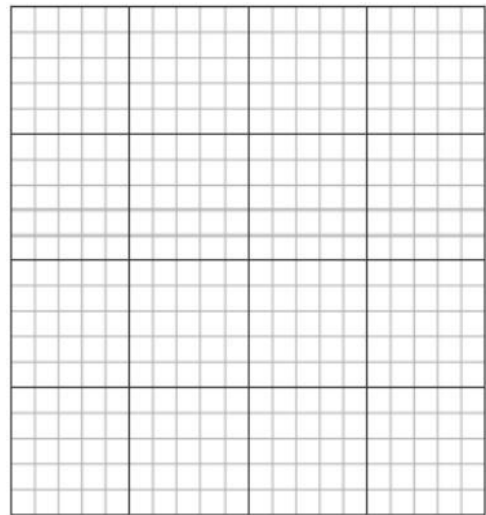


Figure 6 Draw 7V Sinusoidal

Experiment # 2

SIGNAL GENERATOR AND OSCILLOSCOPE

1. Use the signal generator and the oscilloscope to perform the following tasks.
2. Build circuit 'B' shown below in Figure 7.
3. Set the signal generator to generate a sinusoidal signal with a frequency of 200 Hz and peak-to-peak voltage of 7V.
4. Set up one probe across input signal, and another across R1, R2 and R3 alternatively.
5. Measure and draw all the voltages and time related parameters on the graphs below.

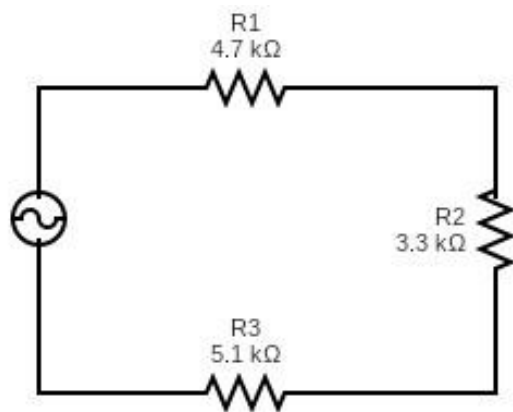


Figure 7 Circuit 'B'

Resistor	Input voltage	RMS Voltage	P-P Voltage
R1	7	0.89	2.51
R2	7	0.62	1.76
R3	7	0.96	2..73
R1 and R2	7	1.51	4.27
R2 and R3	7	1.58	4.49

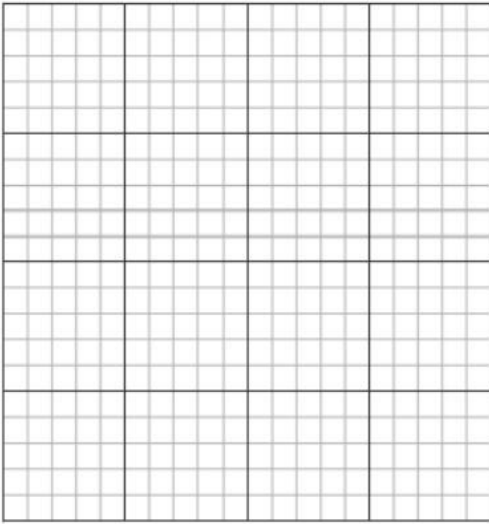


Figure 8 Input Voltage Signal

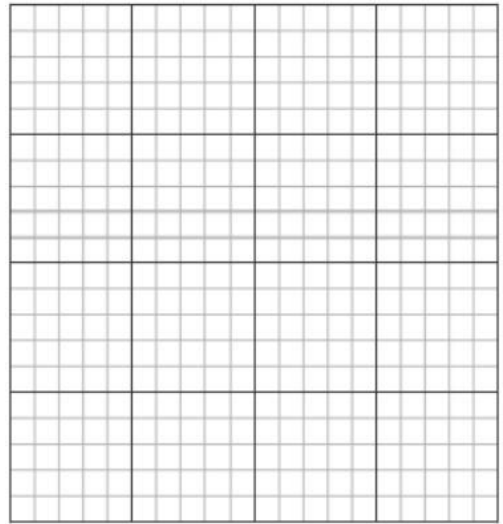


Figure 9 Voltage Signal across R1

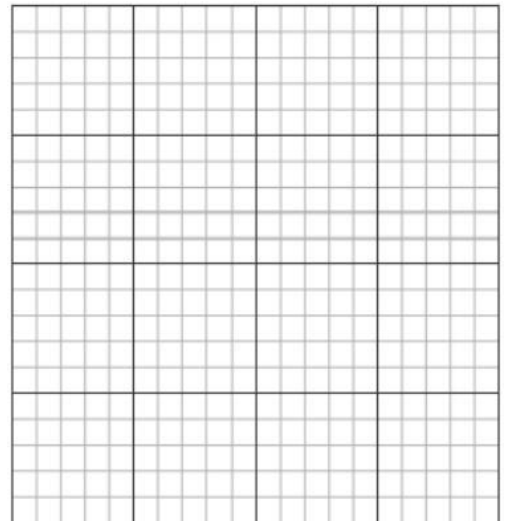
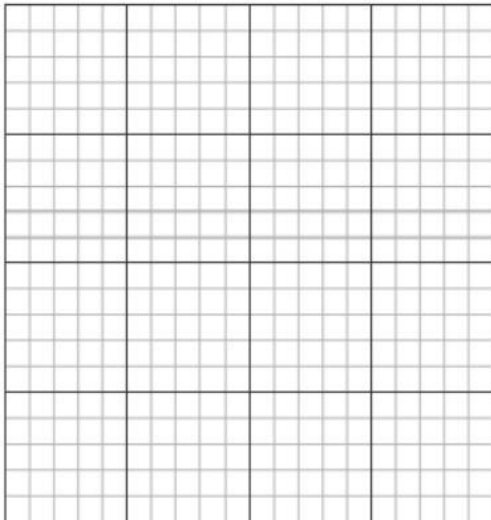


Figure 10 Voltage Signal across R2

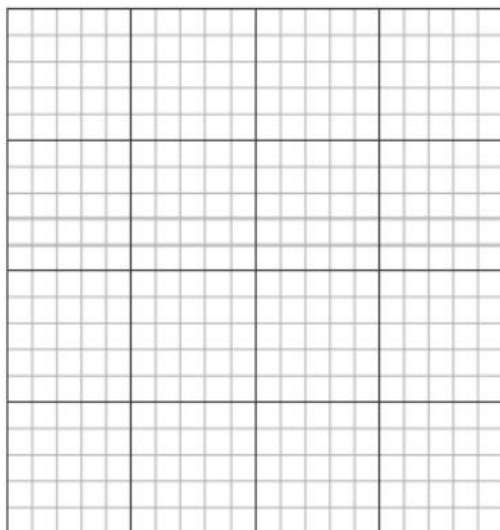


Figure 11 Voltage Signal across R3

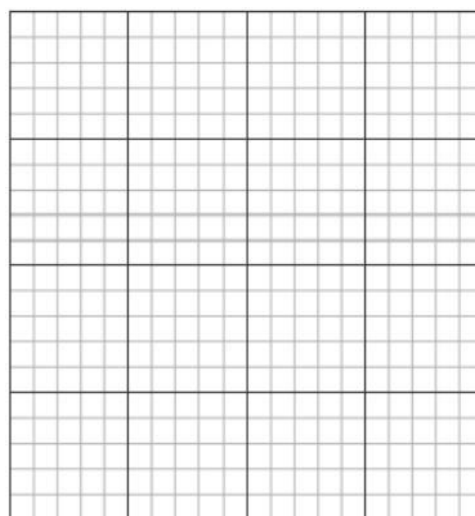


Figure 12 Voltage Signal across R1 and R2

Figure 13 Voltage Signal across R2 and R3

Post Lab Questions

1. Define Ohm's law.

Ohm's Law states that the **current flowing through a conductor is directly proportional to the voltage across it, provided the temperature remains constant**. The relationship is given by the formula $V = IR$, where **V** is voltage, **I** is current, and **R** is resistance.

2. What is resistor?

A **Resistor** is a passive electronic component that **opposes the flow of electric current** in a circuit. It is used to **control current, divide voltage, and protect components**. The resistance value of a resistor is measured in **ohms**.

3. How resistor works?

A **Resistor** works by **opposing the flow of electric current** through it. When voltage is applied across its terminals, it **converts some electrical energy into heat**, limiting the current according to **Ohm's Law** ($I = V/R$). This helps **control current, divide voltage, and protect other components** in a circuit.

4. What is your inference from the input and output waveforms on oscilloscope?

From the **input and output waveforms on an oscilloscope**, we can infer:

1. **Signal Amplification or Attenuation** – If the output waveform is larger or smaller than the input, the circuit is amplifying or reducing the signal.
2. **Phase Shift** – Any horizontal shift between input and output shows a **phase difference**.

3. **Distortion** – Changes in waveform shape indicate **non-linear behavior or distortion** in the circuit.
 4. **Frequency Changes** – Differences in the number of cycles show if the **frequency is altered**.
 5. **Time Delay** – The output may lag or lead the input, indicating **signal delay** in the circuit.
5. Show calculations of 7V input signal (Figure 10) from signal generator.
- a. Voltage calculations (Following volt/dev and number of boxes of oscilloscope)
 - b. Frequency/Period calculations (Following Steps in frequency measurements section)
 - c. Current calculation (Ohm Law)

Solution:

a) Voltage is calculated:

$$\text{Volt/Div} = 1\text{v/div}$$

$$\text{No of divisions} = 7$$

$$\text{Voltage} = 1 * 7 = 7\text{V}$$

b) Time period is calculated:

$$\text{Time/Div} = 0.2\text{ms/div}$$

$$\text{Boxes} = 5$$

$$T = 0.2 * 5 = 1 \text{ ms}$$

Frequency is calculated:

$$F = 1 / 0.001 = 1000 \text{ Hz} = 1 \text{ kHz}$$

c) Current is calculated:

$$I = 7 / 1000 = 0.007 \text{ A} = 7 \text{ mA}$$